SAFARI Stage 1: which bands, which camera, which model

Interim report on the band design study for the SAFARI fuel-condition CubeSat imager, and the experiments behind each choice

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Status: interim. The band, camera and model choices below are the current reading of the evidence, not the final P12 decision. Several runs that bear on them are in progress (section 7). Each conclusion is marked as **established** (a registered prediction tested on complete runs) or **provisional**. Every fire-danger number is our reconstruction of CFFDRS unless it is marked **ISSUED by CWFIS**. The frozen test sets have not been opened: they are read once, in P10, after the decisions are made.

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1. Summary: the current answers

Decision	Current answer	How settled	Main evidence
Bands	VNIR: red (665), red edge (705, 740), narrow NIR (865), with green and blue. SWIR1 near 1600 nm. SWIR2 (2190) helps, but no Dragonfly mode reaches it.	Provisional: the regions are firm; exact sets wait for the E-46 re-scores.	eco_v3 and hs_v4 searches; ground plots and lidar; snow-off (E-60); consensus (Q38)
Camera	Dragonfly Mantis (1U, TRL 9): the MS mode with a custom filter block (490, 560, 665, 705, 740, 865 nm) plus the SWIR mode (4 of DFA's filters), imaging together on one head.	Provisional: E-66 (each Mantis mode scored on its real bands) is ready to run.	A-18 and A-20 (one head, shared aperture); hs_v4: Mantis MS + SWIR ties the best measured set
Model	Onboard: the four-head ecology network (percent conifer, closure, height, fuel family). On the ground: the CFFDRS-structured fire model with fresh weather.	The split is established (E-20). The network size waits for E-47.	E-20: weather 3 days old costs 0.055 AUC between patches; eco_v3
Hardware	Q8 (Zynq UltraScale+) with its DPU runs every size within budget. The Q7 with FINN is possible if its buffers fit once sized. hls4ml is ruled out.	Provisional: power figures and FINN buffer sizing are owed.	E-48 conversion check (P12/p12_convert)

Five things the evidence says

- **SWIR earns its place.** On Sentinel-2 the first band any search picks is SWIR2 (B12) in 11 of 15 runs. On ground plots, which owe nothing to Landsat, SWIR1 lowers percent-conifer error by 1.5 % and height error by 0.33 m. SWIR1 also improves snow-off timing by about 4 days.
- **Red edge is in every best set** and leads percent conifer, though dropping it at six bands costs under 5 % (E-62 refuted on its second half).
- **About 5-8 bands is the range that matters.** Skill rises steeply to two bands and flattens; the best sets hold 7-8, and 5-6 sit about 0.008 below.
- **Within the CFFDRS structure, imagery adds little to fire-danger ranking at 99 fires** (about +0.01 within patches, not separable from zero). Its clearest value is mapping fuel structure, curing on grass, and keeping the inputs current.
- **Fresh weather matters more than anything imagery adds**, so the fire model belongs on the ground and the structure maps onboard.

2. What the study does and how it decides

SAFARI is a CubeSat imager for wildland fuel condition, feeding Canadian fire danger rating (CFFDRS). Stage 1 decides which spectral bands the camera should carry. Rather than ranking bands by how well they predict fire on their own, the study mirrors CFFDRS and asks, input by input, what an imagery estimate would change. The places where imagery can enter the CFFDRS equations are called slots.

Slot (CFFDRS input)	Feeds	Operational source	Imagery estimate
Fuel type	ignition, spread, intensity	CWFIS FBP layer or SCANFI rules	ecology network, fuel family head
Percent conifer (M-1/M-2)	spread, intensity	inventory	percent-conifer head
Canopy closure	ignition (in-stand moisture)	fuel-type default	closure head
Crown base height	intensity (crown fire)	FBP defaults	height head x ratio
Curing (O-1 grass)	ignition, spread	CWFIS NDVI product or default	Sentinel-2 curing
Green-up	ignition, spread	CWFIS green-up product	Sentinel-2 phenology
Foliar moisture (FMC)	intensity	FBP equation (date, place)	LFMC retrieval (not yet usable)
Start-up (snow-off)	all moisture codes	3 days after snow-off, ERA5	Sentinel-2 snow-off date

Where imagery can enter CFFDRS. The full table, with the equations each slot feeds, is in docs/CFFDRS_CHAIN.md.

Data

Data	What it is used for
Sentinel-2 L2A composites, 904 tiles (ecology_v3)	band searches for the structure estimators, against SCANFI labels
AVIRIS-NG, 42 lines over boreal Canada (aviris_v4)	band placement anywhere in 400-2500 nm, and each camera's real bands
MAGPlot ground plots, ATL08 lidar	labels that owe nothing to Landsat, to check SCANFI-based rankings
99 fire patches (NBAC), ERA5-Land weather	Stage A: what each CFFDRS input slot is worth for burned area
NFDB lightning starts with matched controls	Stage C: ignition, with weather held fixed within each set
Globe-LFMC (2017+), PROSPECT-D + 4SAIL	live fuel moisture: empirical and simulation-trained retrievals

Rules every experiment follows

- Nested band sets are ranked, never single bands, and each set is judged by training the retrieval model on it (wrapper evaluation).
- Candidate spectra come only from radiative transfer or real hyperspectral data; no multispectral sensor's bands are interpolated into new ones.
- Labels are physical or independently measured, never CFFDRS outputs.
- Train and test are split by place (1-degree blocks), never by random rows. Selection happens inside training folds, and the frozen test sets are untouched.
- Every evaluation includes the sensor noise model (a placeholder until the instrument's SNR is known).
- No band sits in an atmospheric absorption region (rule 7).
- Predictions are registered before results, with what would refute them.

How choices are made

- Performance decides; cost is shown beside each option but never ranks (A-41). Each P12 decision is a table of options with pros and cons, and James chooses (A-49).
- The four structure targets weigh equally. Six other weightings were tried; under five of them the choice did not change (P12/p12_weights).

- Ties are paired: same fold and seed, one standard error of the difference.
- A band set counts as stable when every spectral region in it is chosen by at least 10 of the 15 searches (5 folds x 3 seeds), and the exact consensus set is then re-scored (Q38).

3. What imagery adds to CFFDRS

Stage A trains a CFFDRS-structured network on the development fire patches (99 fire patches in all; a frozen share waits for P10) and asks what each slot is worth when its value comes from imagery instead of the operational source. Scores are AUC for burned pixels: within a patch (which pixels burn) and between patches (which patches burn). 300 fits: 5 folds x 3 seeds.

Rung	Within-patch AUC	Between-patch AUC
R0: CFFDRS physics as is (HFI)	0.669	0.858
R1: learned on the CFFDRS inputs	0.719	0.886
R2: physics + learned residual	0.695	0.880
R4: R2 + a free imagery branch	0.704	0.881
FWI ISSUED by CWFIS (D-1)	no 2 km counterpart	0.857

The P7 ladder (P7/p7_slots). Between patches, none of our rungs separates from the FWI CWFIS issued.

- **Learning on the CFFDRS inputs beats the physics** (R1 - R0 +0.051 within patches, all five folds). Adding a free imagery branch on top adds about +0.01, which 83 fires cannot separate from zero. **Established.**
- **Single slots move little.** Percent conifer and crown base height from imagery change within-patch AUC by about 0.001. Curing from imagery helps on O-1 grass pixels (+0.013, CI above 0), the one slot with a clear gain.
- **Weather age is the largest effect measured** (E-20, below).

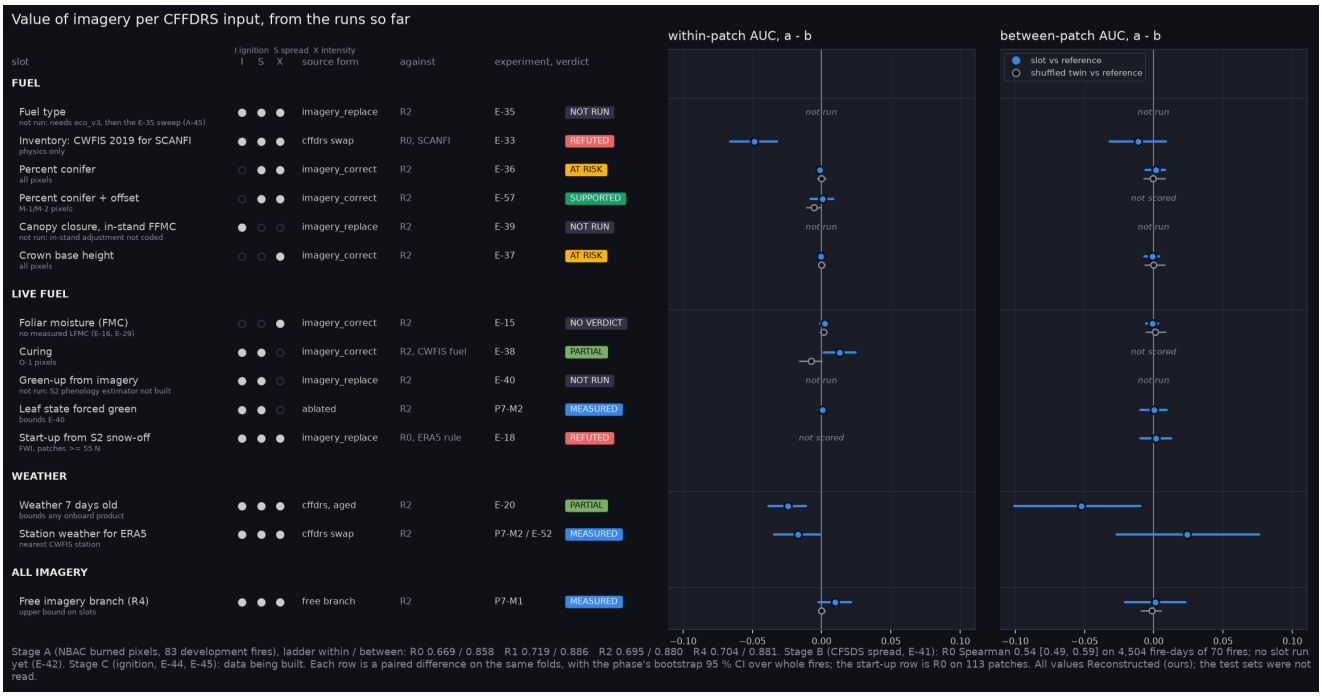


Figure 1. The value map: what each slot changed, by CFFDRS process. Source: docs/results/figures/value_map.png

Weather age (E-20)

One trained model, scored with all its weather 1, 2, 3 and 7 days old. Between patches, R2 loses 0.023 at 2 days (CI reaches 0), 0.055 at 3 days and 0.051 at 7. The physics alone loses more. Within patches nothing changes, because weather is one value per patch. Training on 7-day-old weather is worse at every age. **Established for the decay (PARTIAL: the per-code half was not run).** This is why the fire model is placed on the ground (section 6).

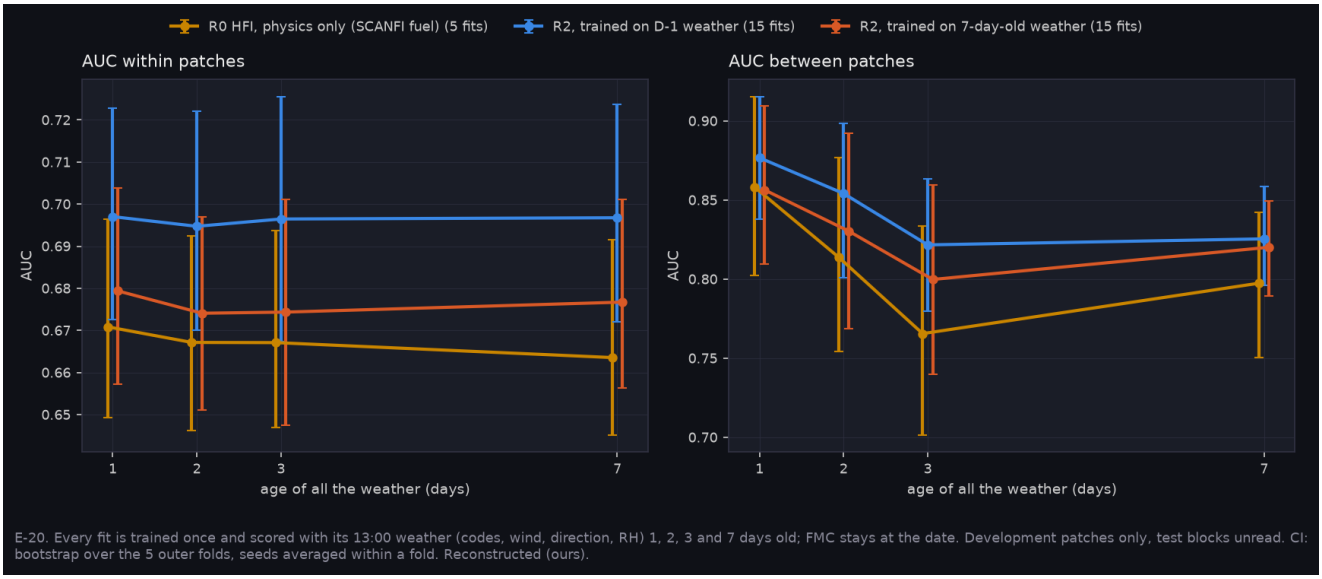


Figure 2. Skill against the age of all the weather (E-20). Source: docs/results/P7/p7d_weather_age/weather_age.png

Ignition (E-44, E-45, partial)

NFDB lightning starts, each matched with up to five controls on the same day under the same weather. CFFDRS physics ranks the start above its controls at 0.533. Learning on the CFFDRS inputs alone (fuel, structure slots, leaf state) reaches 0.586 (+0.052 [+0.033, +0.072]). The imagery rungs are still running, so whether imagery adds to ignition is not yet known. **Provisional (36 of 100 fits).**

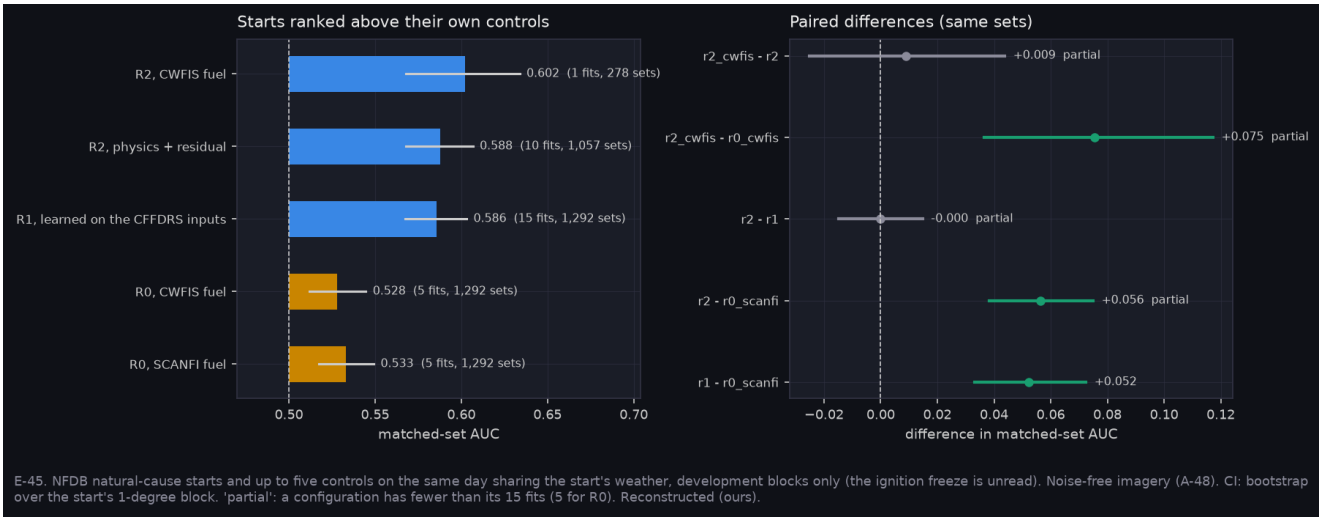


Figure 3. Ignition: matched-set AUC per rung, and paired differences. Source: docs/results/P9/p9b_ignition_rungs/ignition_rungs.png

4. Which bands: the evidence

4.1 Sentinel-2 band searches (eco_v3)

Greedy nested searches over Sentinel-2's 12 bands, inside each training fold, for the four structure targets: percent conifer, closure, height and fuel family (SCANFI labels). 15 searches per selector, with sensor noise.

- **SWIR comes first** (E-22 supported): B12 enters at step 1 in 11 of 15 runs. Closure opens with B12 in all 15; percent conifer with red edge (B7 47 %, B6 27 %); fuel family with NIR.
- **Indices barely help as inputs** (E-59 mixed): raw + indices beats raw only at 8 bands, by 0.003.
- **Fuel family is the weak head:** conifer recall 0.79, deciduous 0.38, mixedwood 0.30. Deciduous and mixed stands are mostly called conifer.

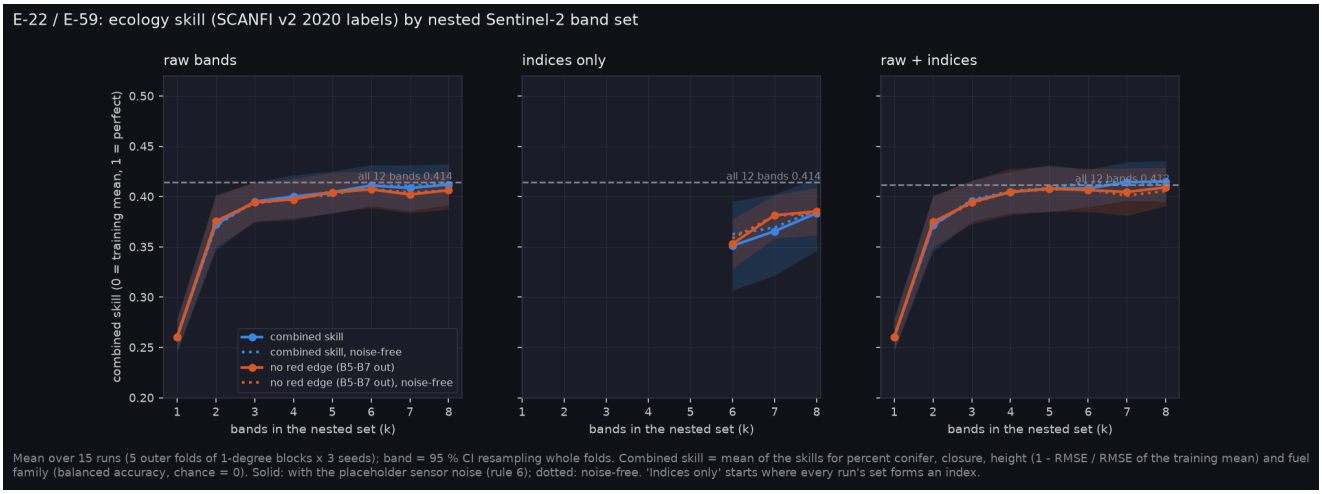


Figure 4. Sentinel-2: skill by nested band count, per search. Source: docs/results/P6/p6a_ecology_v3/curves.png

4.2 Labels that owe nothing to Landsat

SCANFI is mapped from Landsat, SWIR included, so SWIR ranking first could be the labels favouring their own bands. On MAGPlot ground plots, SWIR1 still lowers percent-conifer error by 1.51 % [1.25, 1.70] and height error by 0.33 m, in 5 of 5 folds; NIR is picked first and 1610 nm second. On ATL08 lidar height, adding 2190 nm to 1610 nm helps (-0.07 m). **Established: SWIR's value is not a label artefact.**

4.3 AVIRIS-NG: where in the spectrum (hs_v4)

Real hyperspectral data lets bands be placed anywhere in 400-2500 nm. Searches with the six VNIR bands fixed, and fixed reference sets, all paired with the fixed VNIR on the same folds:

Band set (AVIRIS-NG, hs_v4)	Combined skill	Above fixed VNIR [95 % CI]
Config C: 1610 + 2190 nm (extended InGaAs)	0.288	+0.016 [-0.003, +0.042]
Searched SWIR, extended InGaAs, 4 bands	0.287	+0.015 [+0.004, +0.026]
Sentinel-2, all 12 bands	0.286	+0.014 [-0.003, +0.031]
Mantis MS + SWIR (1240, 1550, 1610, 1680)	0.285	+0.013 [+0.003, +0.026]
Searched SWIR, standard InGaAs, 4 bands	0.279	+0.007 [-0.002, +0.015]
Config A: 1610 nm only	0.278	+0.006 [-0.005, +0.017]
Fixed six VNIR bands	0.272	-

hs_v4 (P6/p6a_hs_v4). Twelve options tie within one SE of the best.

- SWIR improves closure in every set (CIs below 0) and height on both SCANFI and lidar. It does not help percent conifer on SCANFI labels, although it does on ground plots; that remains an open puzzle.

- The searches agree on broad SWIR windows (about 1500-1650 nm, and 2000-2250 nm on extended InGaAs), not on a 50 nm placement.
- The gains are small in combined skill (at most +0.016) but consistent.

4.4 Snow-off and live fuel moisture

- **Snow-off (E-60):** from VNIR alone the start-up date is off by a median 19.5 days; allowing SWIR brings it to 15.7. SWIR1 buys about 4 days, counting toward the SWIR chain. **Established.**
- **Live fuel moisture (E-64):** on conifers neither a simulation-trained nor an empirical Sentinel-2 retrieval beats FBP's own foliar moisture equation; the simulation route is biased +31 points. The forward model's code was checked against Féret's R reference and agrees to 5e-15, so the bias is not a code error. A retrain with priors narrowed inside each fold (E-65) is running. **Provisional: moisture bands are not yet a requirement.**

4.5 Weighting and stability

Re-scoring the searched sets under six weightings of the four targets barely moves the choice: under five of six, the equal-weight search's best set ties with the overall best. The 15 searches rarely agree on a whole set, so the sets were counted by spectral region (Q38). Two Sentinel-2 consensus sets are stable:

k	Sentinel-2 consensus set	Weakest region	Stable
1	B12	11 / 15	yes
4	B4, B6, B8A, B12	9 / 15	no
5	B4, B5, B8A, B11, B12	10 / 15	yes
6	B4, B5, B6, B8A, B11, B12	8 / 15	no
8	B3, B4, B5, B6, B8, B8A, B11, B12	9 / 15	no

Consensus by region (P12/p12_bands). Each set is being re-scored as a fixed set on the same folds (E-46).

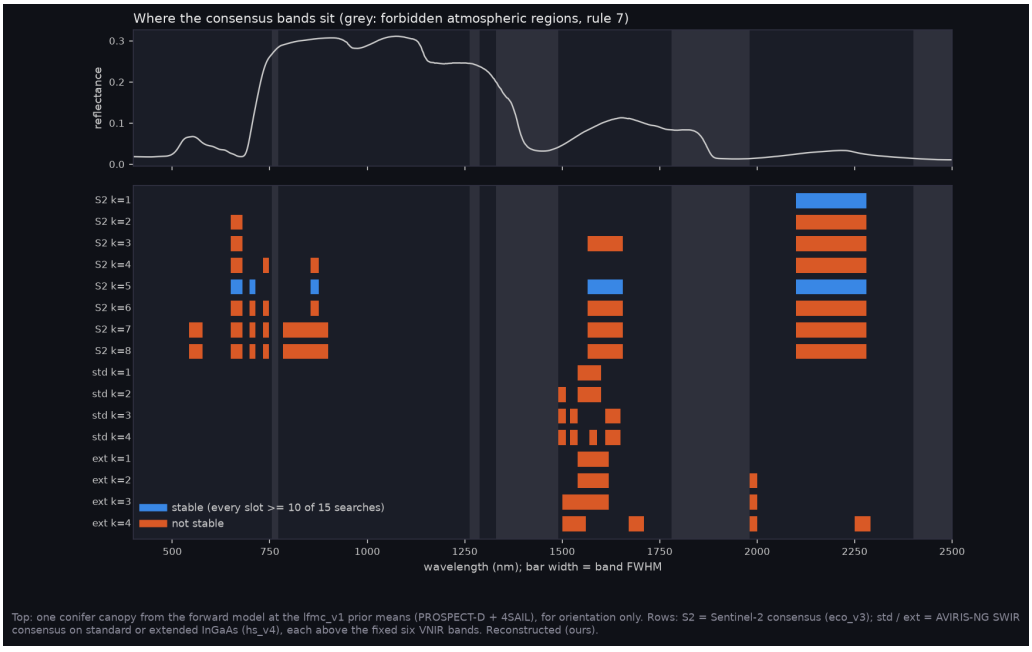


Figure 5. Where each consensus set's bands sit, against a conifer canopy spectrum; grey: forbidden atmospheric regions. Source: docs/results/P12/p12_bands/qual_E-46_band_placement.png

4.6 Band conclusion (provisional)

Carry **red, red edge (705 and 740), narrow NIR (865), green and blue** on the VNIR side and **SWIR1 near 1600 nm**. SWIR2 (2190) adds height and closure skill, but no Dragonfly mode reaches it (A-18), so it is out unless a different SWIR camera is chosen. The exact set and band count follow the E-46 re-scores. The front-runners are the stable five-band Sentinel-2 set (red, red edge, narrow NIR, SWIR1, SWIR2; on Mantis without SWIR2) and the 7-8-band sets.

5. Which camera

The inventory (CubeSat_Sensors_and_Lenses.xlsx) and Dragonfly Aerospace's published data set the options. Two facts dominate: no Dragonfly SWIR mode reaches SWIR2 (it stops at 1750 nm, with about 1700 nm usable on standard InGaAs, A-19), and Mantis carries its MS and SWIR modes on one head, imaging at the same instant through a shared aperture with a dichroic (A-18, A-20, confirmed). Chameleon and Caiman are out of scope (James, 2026-11-06).

Mantis mode	GSD at 500 km	Swath	Bands	Bit depth
RGB snapshot	16 m	32 km	standard RGB	8 or 10
Multispectral pushbroom	16 m pan, 32 m MS	32 km	6 (standard: 480, 545, 660, pan 625, 780, 900)	10 or 12
Hyperspectral pushbroom	32 m	32 km	96, 470-900 nm, chosen in orbit	8 or 10
SWIR pushbroom	as MS (A-20)	32 km	4 of 12 DFA filters, 1000-1750 nm	not stated

Dragonfly Mantis datasheet and band tables (config/vendors/dragonfly.yaml, A-52). Every mode: 1U, under 0.5 kg, under 6 W imaging, TRL 9. No SNR is published.

DFA's standard MS block is not the block the evidence asks for: it has no red edge and no 865 nm, and two of its bands (pan, 780 nm) overlap the oxygen absorption at 760 nm. DFA builds filter blocks in-house, so the custom block (490, 560, 665, 705, 740, 865 nm) is orderable. Of DFA's 12 SWIR filters, 10 are usable: 1348 nm sits in water vapour and 1702/87 runs past the standard-InGaAs edge.

Options

Option	Pros	Cons
A. Mantis MS (custom block) + SWIR (current lean)	Ties the best measured set on AVIRIS-NG (0.285 vs 0.288). The custom block carries the red edge and 865 nm the searches want. One head, one aperture: SWIR and VNIR co-registered by design. 1U, under 6 W, TRL 9.	No SWIR2. Custom filters must be ordered (DFA builds them in-house). SNR not published. The 1059/1068 nm filters sit near the dichroic crossover.
B. Mantis MS (standard block) + SWIR	Off the shelf.	No red edge or 865 nm; pan and 780 nm overlap the O2-A absorption (rule 7, scored as published by James's decision).
C. Mantis HS + SWIR	Any VNIR set can be chosen in orbit, a hedge on the band decision.	470-900 nm loses 443 nm (aerosol) and 945 nm; 96 bands raise downlink and onboard compute; untested against 6 good bands (E-66 will test it).
D. Mantis MS only	Simplest; VNIR alone reaches about 94 % of the best measured score.	Loses SWIR's closure, height and snow-off gains.

E-66: each Mantis mode on its real bands

Approved and ready to run on AVIRIS-NG, on the same folds as hs_v4: the RGB proxy, the standard MS block as published and its clean subset, the custom block, all 81 HS surface channels and the best 6 of them, SWIR alone

(best 4 DFA filters), and each MS block with its best 4 filters. Registered predictions: the custom block beats the standard one; MS beats SWIR alone; HS is no better than 6 well-placed bands; SWIR adds to MS; RGB is the floor.

Current lean (provisional): option A, the Mantis MS mode with the custom filter block plus the SWIR mode, on one head. E-66 will show whether the real filters keep it tied with the best, and how much the standard block and the HS mode give up.

6. Which model, and where it runs

Onboard, or on the ground?

The structure estimators (percent conifer, closure, height, fuel family) need no weather, so they can run onboard and downlink small maps. The fire-danger model needs weather, and E-20 shows that weather even 3 days old costs about 0.05 AUC between patches, while training on stale weather does not recover it. A product uplinked from orbit is 1-2 days old at best. **So the fire model runs on the ground with fresh weather, and the onboard model is the ecology network (established split, A-41).**

The onboard model

TinyEcologyUNet: a small U-Net over the chosen bands and their indices, with four heads. The bands were ranked with base 8 channels and depth 2 (about 32,000 parameters). E-47 (registered) trains 12 sizes, base 4-32 channels by depth 1-3, at the chosen bands; the hypothesis is that the ranked size is within one paired SE of the best. It waits for the band set.

Does it fit the flight computer? (E-48 pre-check)

Flow	Converts	Fits at <= 80 % of a board	Seconds per 1000 x 1000 px scene
hls4ml	12 of 12 sizes; C simulation matches to 0.0014	none (the ranked 8x2 size needs 258 % of the Q8's memory)	not measured
DPU on the Q8 (INT8)	12 of 12, each one DPU subgraph	fixed overlay, mostly idle	0.015 at 8x2, at most 0.28
FINN (W4A4 / W8A8)	24 of 24	Q8: 23 of 24; Q7: 20 of 24 (buffers not yet sized)	8.6 at 8x2; 138 at the largest sizes

All 12 sizes, 22 inputs (8 bands with indices), untrained. Budget: 43 s per scene. The index step divides, so it runs as a separate pointwise stage, as the existing ratios front end does (0 memory, 6 % of the Q8's logic).

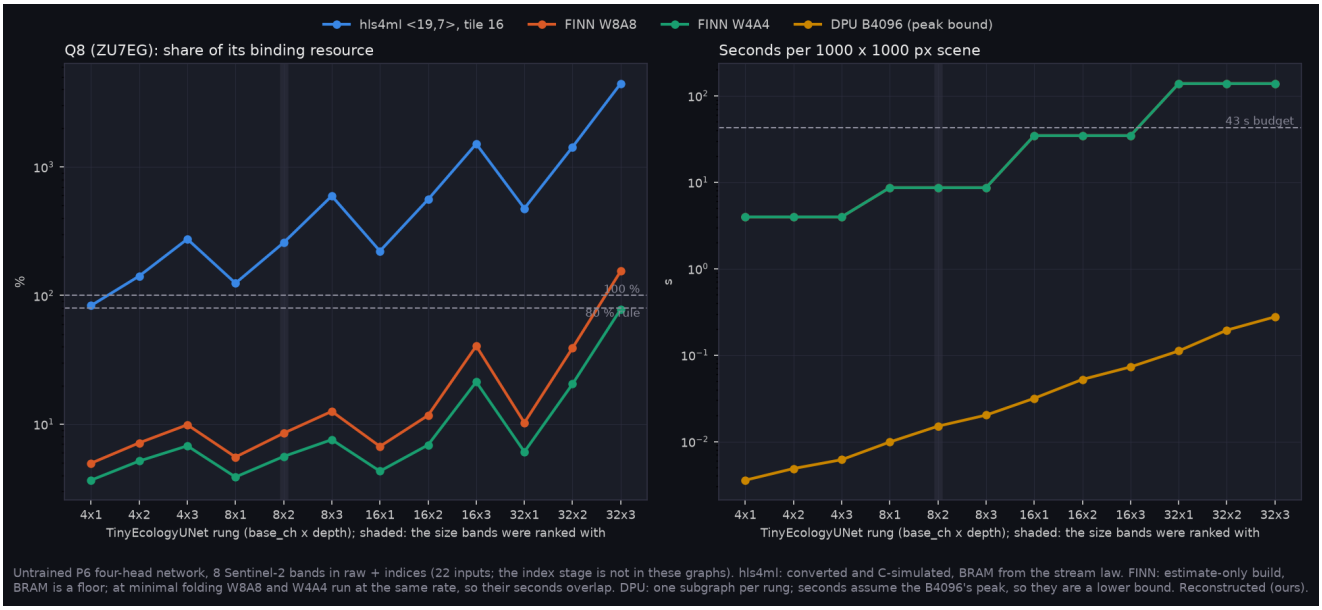


Figure 6. Memory share of the Q8 and seconds per scene, by network size and flow. Source: docs/results/P12/p12_convert/convert.png

Hardware options (provisional): the **Q8 with its DPU** is safe on time for every size (3.5 W typical board, mostly idle); the **Q7 with FINN** (2 W board) fits on FINN's estimates, but only buffer sizing, which needs Vivado on Linux, will confirm it. hls4ml cannot carry a U-Net of useful size on either.

7. What is still running, and what it could change

Run	Question	Could change
E-46 re-scores (Narval)	Each consensus band set scored as a fixed set	the band count and exact set
E-66 sensors_v1 (ready)	What each Mantis mode offers on its real bands	the camera option and its filter block
E-45 ignition, remaining fits	Does imagery add to ignition ranking?	whether imagery carries ignition value
E-65 LFMC v2 (Narval)	Does a right-sized prior make the simulation retrieval work?	whether moisture bands (1240 nm) become a requirement
E-35 fuel slot	Does an imagery fuel map beat the inventory in CFFDRS?	the value of the structure estimators to fire danger
E-47 size ladder	Smallest network within one SE of the best	the onboard model size and hardware fit
E-48 buffer sizing, power	Q7 with FINN, and watts per option	the flight computer

Once these land, P12 writes the three option tables (bands, architecture, hardware) for James to choose from, and P10 scores the chosen configuration once on every frozen test set.

8. Limits

- Most structure labels are SCANFI, which is mapped from Landsat; ground plots and lidar check the main rankings but cover fewer targets.
- Stage A rests on 99 fire patches, 72 of them from 2023; between-patch comparisons have wide intervals.
- Sensor noise is a placeholder model: no Mantis SNR is published, so camera comparisons are spectral, not radiometric.
- GSD and swath are not modelled: training data are at 20 m, Mantis MS at 32 m.
- Globe-LFMC has no Canadian sites; US jack and red pine stand in.
- The turbid-medium canopy model (4SAIL) is a simplification for conifer stands.

Appendix A. The experiment register

Every registered experiment, its state and the latest verdict recorded in a phase sheet. Hypotheses, variables and full results are in docs/EXPERIMENTS.md and the results packages under docs/results/.

ID	Experiment	State	Latest verdict
E-1	VNIR band selection	partial	-
E-2	All-11 greedy	partial	-
E-3	Forced-SWIR delta	partial	-
E-4	SWIR-only floor	runnable	-
E-5	Leave-one-out band value	partial	-
E-6	Selection stability	partial	-
E-7	Resolution requirement	runnable	-
E-8	Label independence control	needs pack	-
E-9	Lightning cohort	needs data	-
E-10	Negative control	partial	-
E-11	Greenness vs moisture indices	runnable	-
E-12	Index baseline vs learned bands	runnable	-
E-13	Weather representation	runnable	-
E-14	Fusion architecture	runnable	-
E-15	Imagery vs the FBP formula	runnable	-
E-16	FMC arms	blocked	-
E-17	Station-density stratification	needs code	-
E-18	Snow-off startup	runnable	REFUTED
E-19	Latitude-banded weather	runnable	-
E-20	Weather staleness	partial	PARTIAL
E-21	Fuel-layer refresh	partial	AT RISK
E-22	Ecology arm	runnable	SUPPORTED
E-23	SWIR band placement	needs data	-
E-24	1209 vs 1240 nm	needs data	-
E-25	Spectral resolution: HS vs MS	needs data	-
E-26	Within-fire burn pattern	runnable	-
E-27	Cost per input channel	runnable	-
E-28	Co-registration tolerance	runnable	-
E-29	T1 live foliar moisture	runnable	SUPPORTED
E-30	T2 curing / NPV fraction	blocked	-
E-31	Fire occurrence prediction	blocked	-
E-32	CFFDRS as is (R0)	partial	SUPPORTED
E-33	Fuel inventory comparison	partial	REFUTED
E-34	Learned CFFDRS (R1/R2)	runnable	MIXED
E-35	Fuel-type slot	runnable	-
E-36	Percent-conifer slot	runnable	AT RISK

ID	Experiment	State	Latest verdict
E-37	Crown-base-height slot	runnable	AT RISK
E-38	Curing slot	runnable	PARTIAL
E-39	Canopy closure to in-stand moisture	needs code	-
E-40	Green-up slot	needs code	-
E-41	Spread R0 (CFSDS)	partial	PARTIAL
E-42	Spread slot value	needs pack	-
E-43	Date-only control	partial	SUPPORTED
E-44	Ignition R0	partial	SUPPORTED
E-45	Ignition slot value	runnable	PENDING
E-46	Band and index decision	runnable	PENDING
E-47	Architecture decision	runnable	-
E-48	Hardware decision	needs pack	-
E-49	Dating the weather (R0)	partial	REFUTED
E-50	Latitude-correct day length (R0)	partial	SUPPORTED
E-51	Operational start-up (R0)	partial	MIXED
E-52	Weather source: station vs ERA5 (R0)	partial	REFUTED
E-53	Weather staleness under R0	partial	SUPPORTED
E-54	Legacy composite leakage	partial	MIXED
E-55	Operational FWI grid vs the reconstruction	runnable	MIXED
E-56	Operational FBP outputs on a 2024-2025 cohort	partial	PARTIAL
E-57	Percent-conifer correction with an imagery-free offset	runnable	SUPPORTED
E-58	Closing R2's within-patch gap to R1	runnable	REFUTED
E-59	Indices as model inputs	runnable	MIXED
E-60	Snow-off without SWIR	runnable	REFUTED
E-61	Best SWIR bands, searched broadly	partial	REFUTED
E-62	Best VNIR bands, searched broadly	partial	REFUTED
E-63	Out-of-time generalisation, two training regimes	partial	AT RISK
E-64	LFMC: simulation-trained, Sentinel-2 empirical, FBP equation	runnable	REFUTED
E-65	LFMC: simulation retrieval with fold-wise narrowed priors (P6f-v2)	runnable	MIXED
E-66	What each Dragonfly Mantis mode would offer	runnable	MIXED

Appendix B. Reproducing this report

- Repositories: Fire_Risk_Statistics (design, analysis, this report), Fire_Risk_Model (data and packs), Fire_Risk_FW (models, cluster runners, FPGA flows).
- Every results package is regenerated by the command at the top of its README; cluster runs are copied to narval_results/.
- Decisions and their reasons: docs/ASSUMPTIONS.md (A-1 to A-52); open questions: docs/OPEN_QUESTIONS.md; the plan: docs/PLAN_NEXT.md.
- This report: python scripts/make_report.py (needs the report extra).